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Early Prediction of Parkinson's Disease using Voice Biomarkers

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Abstract Parkinson's disease is a neurodegenerative disorder. is progressive in nature and generally affects motor functions, including those of speech. Thus becomes an important factor early detection of this can get management and treatment. The project uses machine learning to predict the PD using Voice biomarkers from UCI Parkinson dataset. Such data include a number of acoustic parameters related to jitter: shimmer, and pitch frequency among others which reflect vocal instabilities commonly found in PD patients. The data preprocessing process includes standardization, and then two classification algorithms are trained, Random Forest and Support Vector Machine. By virtue of hyperparameter tuning and cross-validation, the performance of the Random forest model came out to be much better with around 95- percentage accuracy and recall of 97%, hence making it efficient for initial stage detection of the disease. Feature importance analysis revealed that jitter and Shimmer are attributed to be important indicators.

Keywords— Parkinson's Disease, Voice Biomarkers, Machine Learning, Random Forest, Support Vector Machine (SVM), Feature Importance, Early Diagnosis, Speech Analysis, UCI Dataset, Telemedicine.

I. INTRODUCTION

This includes the physical, emotional, social, spiritual, and Financial burden of taking care of the patients with PD. The review describes some of the unique features of Parkinson's, including the following: Motor and non-motor changes often lead to increases in patient needs and dependence on caregivers. Authors synthesis evidence about variables related to patient and caregiver with increased burden, including: symptom severity, time spent caregiving and financial stress. They further discuss negative consequences to the health and quality of the caregivers, such as heightened anxiety and susceptibility to depression. Further in the paper, the interventions to decrease burden to caregivers and update to help clinicians identify and Address this problem in post pandemic era [1]. This Study explores how motor rhythm disturbances are evident in PD patients and hypothesises that dysrhythmia is not uncommon. and can easily be identified using digital techniques. Participants completed the tapping tasks via a special smartphone program that permitted quantitative analysis of the timing of the movement and the coordination of its movement. Authors reported that PD we found prominent rhythmicity deficits in patients compared with healthy controls in supporting the dysrhythmia as a distinctive and quantifiable motor symptoms. An App-based test that can provide a scalable, objective method for screening for and monitoring motor dysfunction in Parkinson's disease [2].

The proposed method is a systematic feature selection framework for selecting the most discriminating features from biomedical data and thus helps in increasing the classification performance by decreasing the computation complexity. It filters remove redundant and irrelevant features at different stages beforehand. Training the model. The authors have proved their method has much better detection performance in comparison of the state-of-the-art methods and is robust even when the amount of data is limited. Various experiments on various data and classifiers. The method has been proven and found suitable for practical clinical application. The proposed framework makes possible the early detection of Parkinson's disease-in an objective and scalable way. With this work, it is established that intelligent feature. Selection is an important driving force in the development of. Diagnostic medical technologies [3].

This project is a development of end-to-End machine learning. Workflow of early detection of Parkinson's using UCI repository of voice biomarkers. The process begins by Performing thorough exploratory data analyses that it includes features inspection, class distribu i correlation studies. It is followed by Normalization, Missing

Value Handling and the irrelevant columns are removed for data consistency and preparation for further analysis. The data was divided into training and test stratified sampling which separates the data and maintains the class balance in the original data set. Both the Random Forest and Support Vector Machine were trained. and whose hyperparameters are optimized by a grid search with cross validation with a focus on recall to maximize The identification of cases having Parkinson's disease. Finally, the model performances were tested based on standard models: suitable visualizations in terms of confusion matrices and ROC curves along with feature importances analysis of key predictors including jitter and shimmer. Finally the pipeline was further validated by making predictions on new samples, and deliver not only class labels, but probability-based confidence way. The model that is trained and the pre-processing tools are saved, and they can be easily integrated in future applications for digital health screening and telemedicine [4].

II. LITERATURE SURVEY

[5] It should be Evolved into the widely accepted eponym *Parkinson's disease*, tracing its historical adoption and reflecting on the broader role of eponyms in medical naming.

[6] In societal/contextual issues (including the COVID- 19 pandemic, stigma and limited support) — and that although some interventions show promise, there remains a need for more rigorous, targeted strategies to reduce this burden

[7] Developed a wrist-worn wearable system that continuously records motion and computes two digital biomarkers — motor activity and a rest-tremor index — to help distinguish early Parkinson's disease patients from healthy controls.

[8] It should be detected using a smartphone-based tapping test. PD patients had greater tap-to-tap variability than healthy controls, and tapping speed slightly declined over one year, suggesting potential for digital motor monitoring in PD.

[9] It will study presents a novel multi-level feature selection framework applied to voice-based recordings that, when combined with an optimized KNN classifier, achieves high accuracy (≈95.4%) in detecting Parkinson's disease.

[10] Developed a hybrid method combining optimized learning vector quantization (OLVQ1) and adaptive neuro-fuzzy inference system (ANFIS) that, using vocal feature data, predicts both Motor and Total Unified Parkinson's Disease Rating Scale (UPDRS) scores with excellent accuracy (highest R^2values ~0.99 and lowest RMSE ~0.56.

[11] Developed a multitask deep-neural-network using voice-derived features that jointly classifies whether a patient's Parkinson's disease is severe vs non-severe and regresses their motor and total Unified Parkinson's Disease Rating Scale (UPDRS) scores, achieving exceptionally high accuracy.

[12] Parkinson's disease imposes a significant and increasing economic burden worldwide, with total costs rising sharply as disease severity progresses—mainly due to hospitalization, medication, and productivity losses—highlighting the urgent need for better management strategies and healthcare planning.

[13] This study protocol suggests a multi cohort cross sectional research approach, which includes data collection and analysis of cognitive and behavioral digital biomarkers in people experiencing neurodegenerative diseases. This will lead to an increase in the early detection, monitoring and understanding of the course of neurodegenerative diseases.

[14] Evidence that machine learning has the ability to differentiate between patients motor status with impeccable accuracy is provided when the NNC methodology was applied to a digital biomarker dataset related to Parkinson's diseases to showcase the potential of digital biomarkers to monitor the disease objectively.

[15] Motor testing using a smartphone can be used to effectively discriminate between idiopathic REM sleep behavior disorder in patients, healthy controls, and Parkinson's disease patients. It therefore becomes an example of one of the practical tools for the early detection and monitoring of neurodegenerative motor impairment.

[16] Real-life gait performance measured by wearable sensors are able to be an reliable digital biomarker to detect and monitor motor fluctuations of patients with PD in an out-of-clinic environment.

[17] That digita biomarkers based on smart phones, collected in a remotely administered setting can obtain reliable measures of both motor and non-motor symptoms in patients with PD, thus enabling scalable and accessible disease monitoring.

[18] That wearable sensor-based inspecting can successfully screen for early stage of Parkinson's disease remotely and can offer a practical, non-invasive method of early detection and monitoring of Parkinson's disease.

[19] That the digital biomarkers from the m-Power data has good test retest reliability and longitudinal stability support the potential of using it to monitor the progression of Parkinson's disease in a real-world environment.

[20] That the non-motor features of Parkinson's disease, such as the cognitive, autonomic and psychiatric symptoms, have their own neuropathological correlates underscores the fact that Parkinson's disease is a more global disorder that involves widespread areas of the brain other than the dopaminergic system.

[21] That voice signal analysis can estimate the severity of the disease in Parkinson's, and this

provides a noninvasive tool in monitoring disease progression.

III. METHODOLOGY

The Proposed System Here's a methodology section of a project you would write on the detection of Parkinson's Disease using biomarkers in voice in sentence form. Figure 1: Overall Architecture will go here in the methodology

A. Voice Data Collection

It receives voice recordings from PD patients and healthy people. The recordings of the speech consist of sustained speech of vowels such as /a/, /e/, /i/, and short sentences of speech to record different speech features.

B. Preprocessing

Get rid of background noise and artifacts in recordings; scale audio signals to a conventional amplitude range, and use some trick of smoothing or filter to enhance the sound.

C. Feature Extraction

Extract acoustic features that reflect vocal changes, such as:

- Jitter
- Shimmer
- Harmonic-to-Noise Ratio
- Mel-Frequency Cepstral Coefficients

D. Feature Selection

Analyze extracted features to identify the most significant ones for classification, using either statistical methods or model-based selection; for example, random forest importance, L1-norm SVM. Reduce the dimensions to improve accuracy and reduce computation.

E. Model Traninig And Machine Learning

Segmentation of the dataset into the training dataset and the test dataset, which usually means 80:20, demands stratification for balanced classes. Train the machine learning classifiers Random forest and SVM. Use Grid Search with cross- validation in order to optimize the hyperparameters for better performance.

F. Model Evalution

Evaluate these models with regards to accuracy, precision, recall, F1- score and ROC- AUC. View results- Areas of confusion matrice and ROC.

G. Prediction on New Data

PD detection on new voice samples by means of the trained PD model. Get the predictions as well as the probability/confidence scores.

H. Model Saving and Deployment

Save the trained model and scaler to be used in the future in a web or mobile application. This enables the detection of PD in a non- invasive and early way in reality

I. Summary of Methodology

These are neatly divided into six important stages which are simply explained to become perfectly in Table 1 in order to show the detail manner:

STAGE	DESCRIPTION
Voice Input	In this stage, voice samples are collected from both Parkinson's patients and healthy individuals.
Preprocessing	The recorded voice signals are cleaned to remove background noise and distortions.
Feature Extraction	These features capture the vocal fold irregularities that are common indicators of Parkinson's Disease.
Feature Selection	The extracted features are analyzed to identify the most relevant ones that contribute significantly to the prediction.
Machine Learning Classification	Machine learning models like Random Forest and Support Vector Machine (SVM) are trained using the selected features.
Result and Evaluation	These performance metrics such as accuracy, precision, recall, F1-score.

Table I: Methodology of the Proposed Parkinsons Diseases

IV. IMPLEMENTATION

This project was implemented using Python programming language with the help of several machine learning libraries such as Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib, etc. The process begins with the loading of the Parkinson's dataset from the UCI Machine Learning Repository that contains several acoustic features extracted from voice recordings of healthy individuals and Parkinson's patients. The dataset is explored. to get idea about its structure and missing values in it, and also for seeing feature correlations. Then, some pre-processing was performed by eliminating the columns that were not needed such as "name", ensuring that the missing values were taken care of, and the features were scaled using the Standard Scaler to make the ranges of the features the same. The processed data is then divided into an 80:20 ratio of training and testing data set, respectively, using stratified sampling in order to preserve class balance.

The trained models are then used for evaluation on the test dataset based on different performance metrics: accuracy, precision, recall, F1-score and ROC-AUC. Visualizations like confusion matrices and ROC curves are produced to determine the effectiveness of each of the models. The highest performance was obtained using the Random Forest, making it the best model for early detection

Finally, the best model and scaler are saved using Job lib for future deployment, allowing the system to predict Parkinson's Disease for new voice inputs. This implementation provides a complete and practical pipeline for early and non-invasive diagnosis using voice biomarkers in (Fig 1 it shows the total implementation).

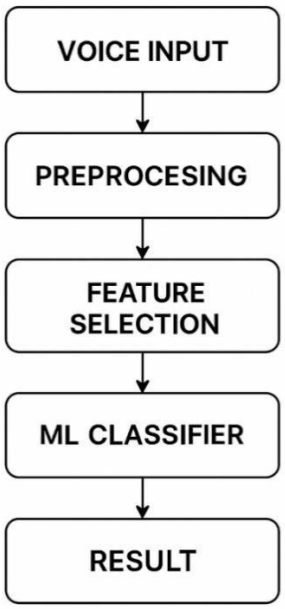

Fig 1: Flow chart of implementation

V. RESULT AND DISCUSSION

The proposed system was successfully implemented using Python and evaluated on the Parkinson's dataset obtained from the UCI Machine Learning Repository. Two machine learning algorithms, Random Forest and Support Vector Machine (SVM), were trained and tested to classify individuals as either healthy or Parkinson's-affected based on extracted voice biomarker

It is evident that the dataset contains a higher number of Parkinson's cases compared to healthy cases. Specifically, there are around 50 healthy samples and approximately 150 Parkinson's samples. This indicates an imbalanced dataset, which is an important consideration for training machine learning models to avoid biased predictions (Fig 2: it shows the healthy and disease while be shown).

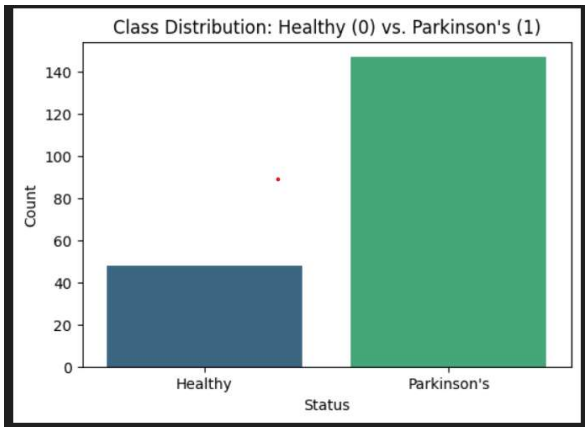

Fig 2: Status Of Healthy Persons And Parkinsons Disease Person

The confusion matrix illustrates the performance of the Random Forest model in classifying individuals as either Healthy or having Parkinson's Disease (PD) While this model made the right diagnosis on 8 Healthy cases and 28 PD cases, it misdiagnosed 2 Healthy people as PD and one PD patients as Healthy. Overall, the model is very accurate for the cases of PD with high accuracy in the correct detection of the cases. Fig 3 it will show the how the confusion matrix of random forest will look like

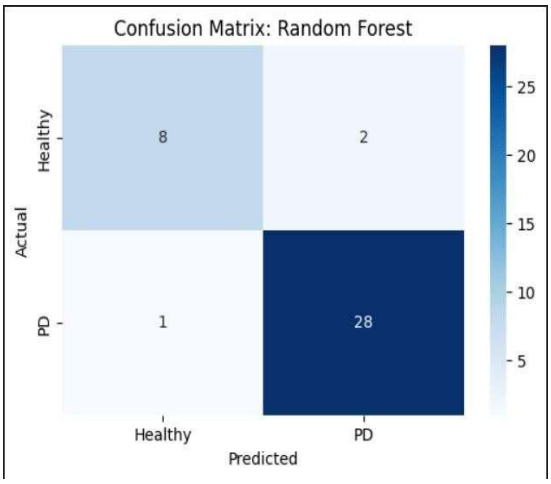

Fig 3: Confusion Matrix of Random Forest

The ROC curve is the performance of the Random Forest model to separate Healthy case and PD case. The curve is high true positive with low false positive, strong classification ability. An AUC value of 0.96 means that the model performance is excellent in differentiating PD patients with Healthy people with high accuracy. Fig 4 will be showing form of random forest curve.

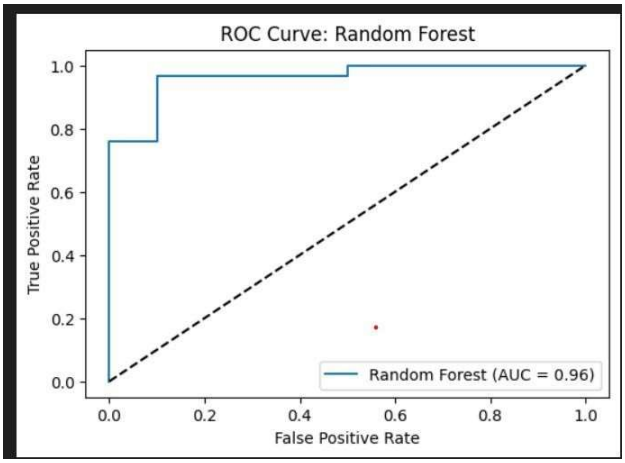

Fig 4: Curve Form

Confusion matrix is used to describe the performance of SVM model in terms of the classification of the subjects as Healthy or has PD. It correctly diagnosed 29 PD cases however all the 10 cases of Healthy people were misdiagnosed with PD, which means that even though we have a SVM model with a high sensitivity of finding PD, our model fails to be specific about identifying Healthy cases. Fig 5 will be shows confusion matrix of SVM

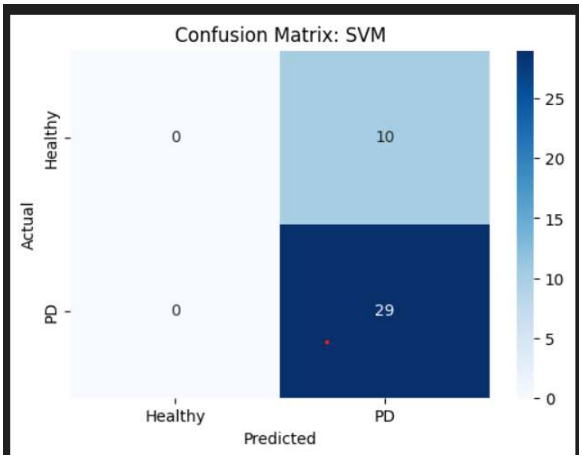

Fig 5: Support Vector Machine

The ROC curve describes the performance of the Support Vector Machine model classification of the Healthy and PD cases. The curve illustrates that the true positive rate is high across most of the false positive rates and is therefore an indicator of a good discriminative ability. The AUC value of 0.92 indicates that the general performance of SVM model is good that can efficiently distinguish between PD and Healthy Indivi. with lesser strength as compared to Random Forest model. Fig 6 will visualize in curve form of SVM.

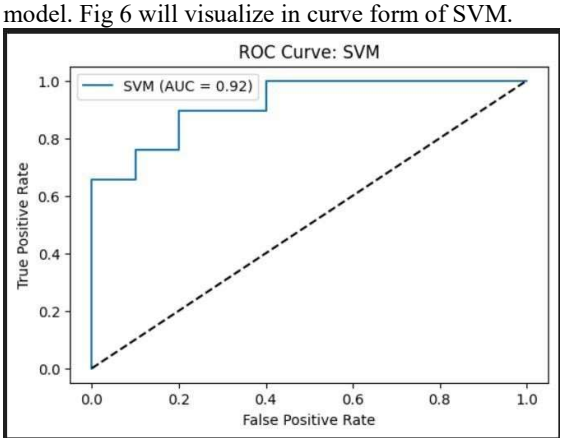

Fig 6: Curve Form

ROC curves of Random Forest and SVM model are given below showing a good comparison between the performance of Random forest and SVM models in discriminating between Healthy and PD patients. It is seen that Random Forest model is playing a better role in terms of classification capabilities with the AUC value which is 0.96 to correctly identify the individuals who has PD and the false positives remains minimal as compared to SVM model. In this regard, the latter, with its AUC of 0.92, also depicts an excellent performance but with a little lower discriminative power. A more steep of a curve the Random Forest has a higher true positive rate at almost every level of the false positive rate. While both models give strong performances, when compared to the overall, Random Forest model provides better predictive accuracy and reliability in the detection of PD. (Fig 7 will be shows in comparison table of Random Forest and Support Vector Machine.

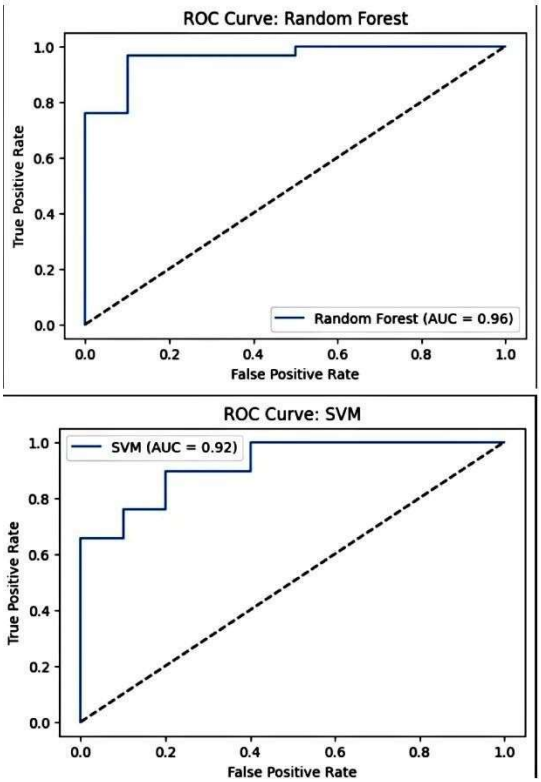

Fig 7: Comparison Table of Support Vector Machine And Random Forest

Fig. below is the top 10 Most informative voice biomarkers using Random Forest model determined to predict PD. Among them, the highest elements of importance are those for PPE and spread1; they have the most contribution to the decision of the model. Other important features, also in the set, are MDVP Fo(Hz), NHR and Jitter:DDP which suggests that variation in pitch and voice frequency is the key in distinguishing the PD patients from Healthy individuals. (Fig 8 will be shows in taking 10 biomarkers in data.

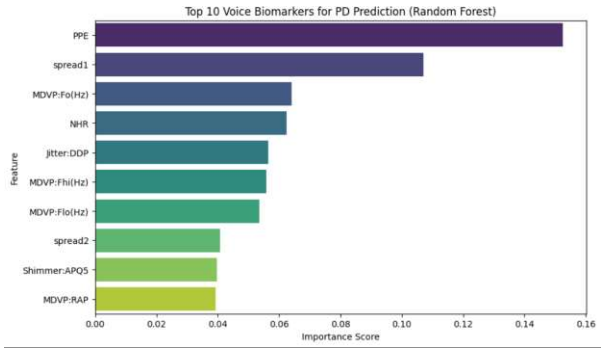

Fig 8: Voice Biomarkers of prediction

VI. CONCLUSION AND FUTURE SCOPE

The presented work in this publication demonstrates a complete pipeline for the early detection of PD using voice recordings. In particular, classical machine learning models especially Random Forest and SVM are very good with a high accuracy and recall for the study, therefore making them viable for screening purposes. Feature analysis has shown that key voice biomarkers, such as jitter, shimmer, and HNR are important in the detection of vocal irregularities t s pipeline implemented including preprocessing, preprocessing,

Feature extraction and feature selection, model training. evaluation, and the pipeline implemented offers is a reproducible and interpretive approach.

In future such deep learning techniques such as CNN, RNN or Transformer-based models may be studied in order to automatically learn some complex features from the voice data to enhance the prediction accuracy. Larger and more diverse datasets including multi-lingual and real-world recordings will lead to more generalization of the models. Integration with mobile and IoT to real-time monitoring can possibly allow early detection of PD using smartphones or wearable devices. Longitudinal studies can also be helpful in determining the course of a disease and the effectiveness of early intervention.

VII. **REFERENCES**

[1] M. Weatherall, 'Parkinson's eponym', Lancet Neurol. vol. 23, no. 2, p. 143, Feb. 2024, doi: 10.1016/S1474-4422(23)00323-X.

[2] W. W. Aamodt et al., 'Caregiver Burden in Parkinson Disease: A Scoping Review of the Literature from 2017-2022', J. Geriatr. Psychiatry Neurol., vol. 37, no. 2, pp. 96–113, Mar. 2024, doi: 10.1177/08919887231195219.

[3] Battista L, Romaniello A. A new wrist-worn tool supporting the diagnosis of parkinsonian motor syndromes. Sensors 2024;24. https://doi.org/10.3390/ s24061965.

[4] Marsili L, Abanto J, Mahajan A, Duque KR, Chinchihualpa Paredes NO, Deraz HA, et al. Dysrhythmia as a prominent feature of Parkinson's disease: an app-based tapping test. J Neurol Sci 2024;463. https://doi.org/10.1016/j.jns.2024.123144.

[5] F. Demir, K. Siddique, M. Alswaitti, K. Demir, and A. Sengur, 'A Simple and Effective Approach Based on a Multi-Level Feature Selection for Automated Parkinson's Disease Detection', J. Pers. Med., vol. 12, no. 1, Art. no. 1, Jan. 2022, doi: 10.3390/jpm12010055.

[6] W. A. Zogaan et al., 'A combined method of optimized learning vector quantization and neuro-fuzzy techniques for predicting uni ed Parkinson's disease rating scale using vocal features', MethodsX, vol. 12, p. 102553, Jun. 2024, doi: 10.1016/j.mex.2024.102553.

[7] M. T. García-Ordás, J. A. Benítez-Andrades, J. Aveleira-Mata, J.-M. Alija-Pérez, and C. Benavides, 'Determining the severity of Parkinson's disease in patients using a multi task neural network', Multimed. Tools Appl., vol. 83, no. 2, pp. 6077–6092, Jan. 2024, doi: 10.1007/s11042-023-14932-x.

[8] Chaudhuri KR, Azulay J-P, Odin P, Lindvall S, Domingos J, Alobaidi A, et al. Economic burden of Parkinson's Disease: a multinational, real-world, cost-of- illness study. Drugs Real World Outcomes 2024. https://doi.org/10.1007/s40801- 023-00410-1.

[9] Crook-Rumsey M, Daniels SJC, Abulikemu S, Lai H, Rapeaux A, Hadjipanayi C, et al. Multicohort cross-sectional study of cognitive and behavioural digital biomarkers in neurodegeneration: the Living Lab Study protocol. BMJ Open 2023; 13:e072094. https://doi.org/10.1136/bmjopen-2023-072094.

[10] Tsoulos IG, Mitsi G, Stavrakoudis A, Papapetropoulos S. Application of machine learning in a parkinson's disease digital biomarker dataset using Neural Network Construction (NNC) methodology discriminates patient motor status. Front ICT 2019;6. https://doi.org/10.3389/fict.2019.00010.

[11] Arora S, Baig F, Lo C, Barber T, Lawton M, Zhan A, et al. Smartphone motor testing to distinguish idiopathic REM sleep behavior disorder, controls, and PD. NEUROLOGY 2018;91:1528https://doi.org/10.1212/WNL.0000000000006366.

[12] Evers LJW, Raykov YP, Krijthe JH, Silva De Lima AL, Badawy R, Claes K, et al. Real-life gait performance as a digital biomarker for motor fluctuations: the Parkinson@Home validation study. J Med Internet Res 2020;22. https://doi.org/ 10.2196/19068.

[13] Goni M, Eickhoff SB, Far MS, Patil KR, Dukart J. Smartphone-based digital biomarkers for Parkinson's disease in a remotely-administered setting. IEEE Access 2022;10:28361–84. https://doi.org/10.1109/ACCESS.2022.3156659

[14] Johnson S, Kantartjis M, Severson J, Dorsey R, Adams J, Kangarloo T, et al. Wearable sensor-based assessments for remotely screening early-stage Parkinson's disease. SENSORS 2024;24. https://doi.org/10.3390/s24175637.

[15] Far MS, Eickhoff SB, Goni M, Dukart J. Exploring test-retest reliability and longitudinal stability of digital biomarkers for parkinson disease in the m-power data set: cohort study. J Med Internet Res 2021;23. https://doi.org/10.2196/ 26608.

[16] Dickson DW, Fujishiro H, Orr C, DelleDonne A, Josephs KA, Frigerio R, et al. Neuropathology of non-motor features of Parkinson disease. Parkinsonism Relat Disord 2009;15(Suppl 3):S1–5. https://doi.org/10.1016/S1353-8020(09)70769-2.

[17] Hemmerling, D., & Wojcik-Pedziwiatr, M. (2022), Prediction and estimation of Parkinson's disease severity based on voice signal, Journal of Voice, 36(3), 439-e9.